

Test Project Module C-CN

Web Technologies

Introduction

简介

Old Shanghai Restaurant requires a mobile- and tablet-first ordering website for dine-in and take-away customers. Dine-in customers identify their table by scanning a QR code with the device camera or by uploading a local QR image. Take-away customers can enter the menu directly. The product is a single-page web application. No page reloads shall occur for any operations.

老上海餐厅需要一个以移动设备和平板设备为优先的点餐网站，供堂食和外带顾客使用。堂食顾客可使用设备摄像头扫描二维码，或上传本地二维码图片来识别餐桌。外带顾客可直接进入菜单。该产品应为单页 Web 应用程序，任何操作均不得引起页面重新加载。

The provided media files include menu data, QR image-reading helper code, table QR images and dish images. Competitors must import the menu data and submit an export of the final database as "db_dump.sql".

提供的媒体文件包括菜单数据、二维码图片读取辅助代码、餐桌二维码图片和菜品图片。选手必须导入菜单数据，并以“db_dump.sql”为文件名提交最终数据库的导出文件。

Description of project and tasks

Database

Import every record from "menu.csv". The following is a reference structure. Competitors may refer to this structure to complete the design of database tables in accordance with the functional requirements of the competition task.

导入“menu.csv”中的每一条记录。以下为参考结构，选手可参照该结构，并根据竞赛任务的功能要求完成数据库表的设计。

dishes

- name
- category
- description
- price
- stock
- image_path

orders

- order_number
- type (Indicates whether the order is dine-in or takeaway)

类型（type，用于指示订单为堂食还是外带）

- total_amount

order_items

- order_id
- dish_id
- dish_name
- unit_price
- quantity
- line_total

Each order has one or more order items. Each order item refers to one dish and must preserve the purchased dish name and unit price as an order snapshot.

每个订单包含一个或多个订单明细。每条订单明细对应一道菜品，并且必须保存购买时的菜品名称和单价，作为订单快照。

Home view

首页视图

- Display three primary buttons: "Dine In", "Take Away" and "My Orders".

显示三个主要按钮：“堂食”、“外带”和“我的订单”。

- After selecting "Dine In", open a drawer and show the "Scan Now" and "Upload QR Code" buttons in the drawer.

选择“堂食”后，打开抽屉，并在抽屉中显示“立即扫描”和“上传二维码”按钮。

- Selecting "Scan Now" must execute code that requests access to the device camera.

选择“立即扫描”后，必须执行请求访问设备摄像头的代码。

- Selecting "Upload QR Code" must open a local image file input and decode the selected image.

选择“上传二维码”后，必须打开本地图片文件选择控件，并解码所选图片。

- A valid table payload uses the format "old-shanghai://table/{number}", where "{number}" is an integer from 1 to 10. A valid result opens the menu and shows that table number.

有效的餐桌数据采用“old-shanghai://table/{number}”格式，其中“{number}”为 1 至 10 的整数。解析结果有效时，打开菜单并显示对应的餐桌号。

- If image decoding fails or the decoded value is not a valid table payload, display exactly "QR code parsing failed" and remain on the home view.

如果图片解码失败，或解码后的值不是有效的餐桌数据，必须准确显示“QR code parsing failed”，并停留在首页视图。

- Selecting "Take Away" opens the menu directly and creates a take-away reference that remains unchanged during refresh. Display the reference with the prefix "Take-away No.".

选择“外带”后，直接打开菜单并创建一个在页面刷新期间保持不变的外带编号。显示该编号时使用前缀“Take-away No.”。

- The takeaway reference shall be a randomly generated 6-digit numeric string and must be unique.

外带编号应为随机生成且唯一的 6 位数字字符串。

Menu view

菜单视图

- The fixed header displays the current table number or take-away reference and a back to home button.

固定页眉显示当前餐桌号或外带编号，以及返回首页按钮。

- Display every imported dish with its image, name, category, unit price, stock status, selected quantity, add control and subtract control.

显示所有已导入的菜品，包括图片、名称、类别、单价、库存状态、已选数量、增加控件和减少控件。

- On mobile devices, each row displays one dish; on tablets, two to three dishes shall be shown per row adaptively based on page width.

在移动设备上，每行显示一道菜品；在平板设备上，应根据页面宽度自适应地每行显示两至三道菜品。

- Do not display the numeric stock value. Display "Out of Stock" when stock is 0, "Low Stock" when stock is from 1 to 9, and "In Stock" when stock is 10 or more.

不得显示具体库存数值。库存为 0 时显示“Out of Stock”，库存为 1 至 9 时显示“Low Stock”，库存为 10 或以上时显示“In Stock”。

- The add control increases quantity by one. The subtract control decreases quantity by one but must never reduce it below 0.

增加控件每次将数量加 1。减少控件每次将数量减 1，但数量不得低于 0。

- A dish with quantity 0 must not exist in the cart.

数量为 0 的菜品不得存在于购物车中。

- The maximum selected quantity for one dish is 10. If stock is below 10, the maximum is the current stock value. A dish with stock 0 cannot be added.

单道菜品的最大选择数量为 10。如果库存少于 10，则最大数量为当前库存值。库存为 0 的菜品不能添加。

- Every quantity change must immediately update the bottom-bar dish quantity and total price.

每次数量变化都必须立即更新底部栏中的菜品总数量和总价。

- The fixed bottom bar displays "Cart", the total dish quantity, the total price and "Checkout".

固定底部栏显示“购物车”、菜品总数量、总价和“结算”。

Cart and persistent state

购物车与持久化状态

- Clicking "Cart" or "Checkout" opens a drawer.

点击“购物车”或“结算”后打开抽屉。

- The drawer displays all selected dishes, showing each item's image, name, unit price, quantity and line subtotal. Below these items are the total price and a "Confirm" button to submit the cart.

抽屉显示所有已选菜品，并展示每项菜品的图片、名称、单价、数量和行小计。在这些菜品下方显示总价和用于提交购物车的“确认”按钮。

- The drawer supports adding, subtracting and deleting dishes. The same quantity and stock limits apply in the menu and drawer. A dish will be removed from the cart when its quantity is reduced to zero.

抽屉支持增加、减少和删除菜品。菜单和抽屉中适用相同的数量与库存限制。当菜品数量减少至 0 时，应将其从购物车中移除。

- The cart drawer contains a "Confirm" button for order submission.

购物车抽屉包含用于提交订单的“确认”按钮。

- Refreshing the menu must restore every cart item, quantity and the current table or take-away reference.

刷新菜单后，必须恢复购物车中的所有菜品、数量以及当前餐桌号或外带编号。

- If the cart is empty, clicking back to home button returns directly to the home view.

如果购物车为空，点击返回首页按钮后直接返回首页视图。

- If the cart contains dishes, clicking back to home button opens a dialog containing exactly "Return to the home page? Unsubmitted cart items will be cleared.". The actions are "Cancel" and "Return Home".

如果购物车中有菜品，点击返回首页按钮后应打开一个对话框，其中必须准确显示“Return to the home page? Unsubmitted cart items will be cleared.”。操作按钮为“Cancel”和“Return Home”。

- Clicking "Return Home" clears all unsubmitted cart and service-context data before opening the home view.

点击“Return Home”后，在打开首页视图之前清除所有未提交的购物车数据和服务场景数据。

Checkout and order processing

结算与订单处理

- Clicking "Confirm" with an empty cart displays exactly "Please select at least one dish" and must not show the result view.

购物车为空时点击“确认”，必须准确显示“Please select at least one dish”，且不得显示结果视图。

- Use the current table or take-away reference and selected dish to create the order.

使用当前餐桌号或外带编号以及已选菜品创建订单。

- Before saving the order, verify that each selected dish exists and that the requested quantity does not exceed the current stock.

保存订单前，验证每道已选菜品是否存在，并确认请求数量未超过当前库存。

- If the stock quantity of any dish added to the cart becomes insufficient when submitting the order, terminate the submission process, show the prompt "Some dishes in your cart are out of stock", and highlight the corresponding dishes. This scenario occurs when two users add the same low-stock dish to their carts simultaneously; if one user completes the order first, the other user will receive this prompt.

提交订单时，如果购物车中任何菜品的库存变得不足，应终止提交过程，显示提示“Some dishes in your cart are out of stock”，并高亮对应菜品。当两名用户同时将同一道低库存菜品加入购物车，且其中一名用户先完成订单时，另一名用户将收到此提示。

- Store the order and its purchased dishes in the database, including an order number and ordered time.

将订单及其已购菜品存储到数据库中，包括订单编号和下单时间。

- After a successful order, reduce each purchased dish's stock by its ordered quantity. Stock must never become negative.

订单成功后，按订购数量减少每道已购菜品的库存。库存不得出现负数。

- After a successful order, clear the stored cart and open the result view.

订单成功后，清除已存储的购物车并打开结果视图。

Result view

结果视图

- Display the table number or take-away reference.

显示餐桌号或外带编号。

- Display the order number and ordered time.

显示订单编号和下单时间。

- Display every purchased dish with image, dish name, unit price and quantity.

显示每道已购菜品的图片、菜品名称、单价和数量。

- Display line subtotals and the total order price.

显示各行小计和订单总价。

My Orders view

我的订单视图

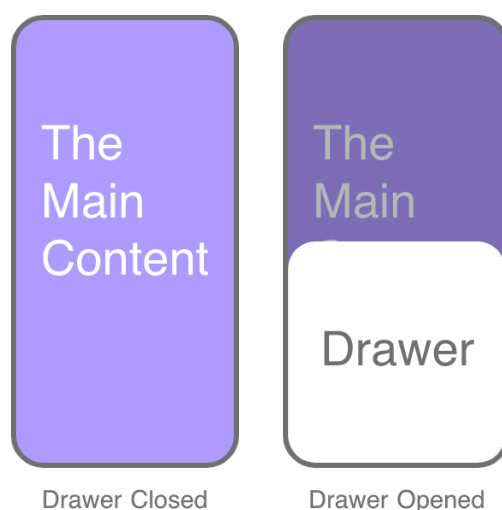
- List all orders submitted by the current device in descending order of ordered time.
按下单时间降序列出当前设备提交的所有订单。
- Each order entry shall display the order number, ordered time and total amount.
每个订单条目应显示订单编号、下单时间和总金额。
- Clicking an order entry opens the result view and displays the detailed data of the corresponding order.
点击订单条目后打开结果视图，并显示对应订单的详细数据。

Drawer Component

抽屉组件

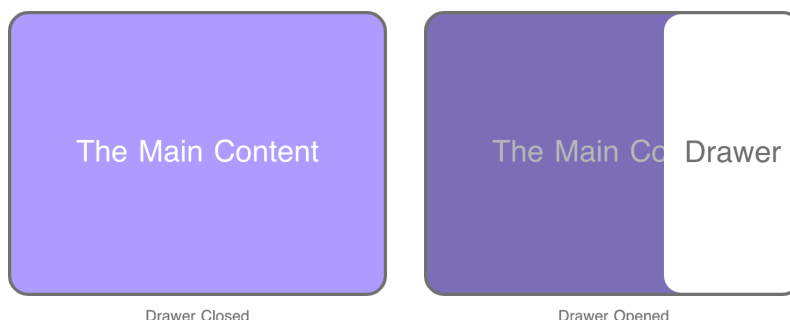
Drawer is a front-end interactive component. Its show and hide animations are slide-in and slide-out transitions originating from the edge of the viewport. When opened on mobile devices, its main content slides upward from the bottom of the viewport. The page content will be covered by the drawer's mask layer. Clicking the mask layer of the drawer closes the drawer component.

抽屉是一种前端交互组件。其显示和隐藏动画为从视口边缘开始的滑入和滑出过渡。在移动设备上打开时，抽屉的主要内容从视口底部向上滑入。页面内容将被抽屉的遮罩层覆盖。点击抽屉的遮罩层可关闭抽屉组件。



On tablet devices, the drawer component shall slide its main content in from the right side of the viewport.

在平板设备上，抽屉组件的主要内容应从视口右侧滑入。



Instructions to the Competitor

选手须知

- Use the supplied menu data. Do not replace the provided dish records with different records.
使用所提供的菜单数据。不得使用其他记录替换所提供的菜品记录。
- Submit the final database export as "db_dump.sql" in the project root. The file must contain imported menu data, a reasonable data structure, and comply with Third Normal Form.
在项目根目录中以“db_dump.sql”为文件名提交最终数据库导出文件。该文件必须包含已导入的菜单数据，采用合理的数据结构，并符合第三范式。
- Use the decodeQrFile method from qr-image-reader.js (parameter: a File object) to parse the QR Code image.
使用 qr-image-reader.js 中的 decodeQrFile 方法（参数为 File 对象）解析二维码图片。
- Home, menu, result, my orders views must change without a full document reload.
首页、菜单、结果和我的订单视图必须在不完整重新加载文档的情况下切换。
- All prices shown to the customer must use exactly two decimal places.
向顾客显示的所有价格必须保留且仅保留两位小数。

Other

其他

The workstation does not provide camera hardware. Competitors only need to implement the code to open and request camera access for the "Scan Now" function. Since physical camera hardware is unavailable, judges will evaluate the relevant source code instead of testing live camera capture. Uploading and decoding a local QR image remains fully assessable.

工作站不提供摄像头硬件。对于“立即扫描”功能，选手只需实现打开摄像头并请求摄像头访问权限的代码。由于没有实体摄像头硬件，裁判将评估相关源代码，而不测试实时摄像头捕获功能。上传并解码本地二维码图片的功能仍将进行完整评测。

Marking Summary

评分汇总

Sub-Criteria 子评分标准	Marks 分数
Database structure 数据库结构	4.1
Home View 首页视图	4
Menu View 菜单视图	9.6
Cart 购物车	4.2
Checkout and order processing 结算与订单处理	4
Result View 结果视图	2.1
My Orders view 我的订单视图	1.2
Application Functionality 应用程序功能	3.8
Total 总计	33.0